Sai Subrahmanyam Mahadasa

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Senior Full Stack Developer with 6 years of experience building workflow-heavy web applications and internal platforms across HR/benefits, financial services, and healthcare. Strong in **React, Next.js, TypeScript** for high-performing UI layers and **C#/.NET Core (ASP.NET Core Web APIs)** for scalable backend services, with **Node.js/Express** used for supporting services, integrations, and async/event-driven patterns. Experienced delivering secure systems with **JWT/OAuth concepts, RBAC**, and audit-friendly logging; optimizing **SQL Server/PostgreSQL** performance; and shipping production code through **Docker + CI/CD** in **AWS/Azure** environments. Recently focused on **GenAI/LLM features** building **RAG-based chatbots** using **LangChain/LlamaIndex**, integrating vector databases, and deploying reliable retrieval pipelines with evaluation and monitoring. Known for strong ownership, clear communication, mentoring, and partnering with product/design/QA to deliver clean UX and predictable systems from design through production support.

TECHNICAL SKILLS

Programming Languages: TypeScript, JavaScript (ES6+), C#, SQL, Node.js, Python (GenAI tooling & scripting)
Frontend (UI Engineering): React, Next.js, TypeScript, React Hooks, component architecture, reusable UI libraries, Redux Toolkit, RTK Query, client-side routing, SSR/CSR concepts, responsive UI (HTML5/CSS3), accessibility fundamentals (WCAG), performance tuning (lazy loading, code splitting, memoization, render optimization), browser profiling/debugging.
Backend (APIs / Services): C#/.NET, .NET Core, ASP.NET Core Web API, RESTful services, GraphQL integrations, layered architecture (controllers/services/data access), Entity Framework Core, LINQ, background processing concepts, Node.js services (Express patterns), async/event-driven workflows, API integrations, DTO contracts, pagination/filtering patterns
Security / Authorization: JWT-based auth patterns (concepts), OAuth 2.0 concepts, RBAC/authorization policies, secure token handling, input validation, safe error handling, audit-friendly logging, PII-safe handling concepts
Databases & Performance: SQL Server (primary), PostgreSQL, MySQL, MongoDB — relational modeling, normalization, stored procedures, indexing strategies, execution plans, query optimization, transactions; Redis (caching/session patterns)
GenAI / LLM / RAG: LLM integration patterns, RAG pipeline design, LangChain, LlamaIndex, embeddings concepts, grounding/citations, evaluation workflows, prompt templates/guardrails, vector DB usage (vector indexing/retrieval concepts), chatbot UX integration
Cloud / DevOps / Delivery: AWS (CloudWatch concepts), Azure, Docker, GitHub Actions CI/CD, Azure DevOps (Repos/Pipelines/Boards), environment-based configuration, monitoring/logging concepts, incident workflows, release support
Testing & Quality: Jest (React), xUnit/MSTest (C#), integration testing patterns, API testing with Postman, code reviews, debugging, maintainability improvements; familiarity with SAST gates (Veracode-style scanning)
Tools & Platforms: Visual Studio, VS Code, Postman, Fiddler, Git/GitHub, Jira, Confluence, ServiceNow, Chrome DevTools

PROFESSIONAL EXPERIENCE

Perky (Perspective Partners LLC , client : Lincoln Finance) — Senior Application Developer (Full Stack)
New York, NY (Remote) | Mar 2024 – Dec 2025
Domain: HR Leave Platform (SaaS) | **Stack:** React, Next.js, TypeScript, Node.js/Express, C#, ASP.NET Core Web API, PostgreSQL, REST/GraphQL, Docker, GitHub Actions, AWS

- Owned end-to-end delivery for leave-management features from clarifying requirements with product/QA to implementation and release so UI and API changes shipped with clearer acceptance criteria and fewer late-cycle surprises.
- Built React + Next.js UI experiences in TypeScript (dashboards, forms, approvals, admin tools) using reusable component patterns and predictable data flow (**Redux Toolkit/RTK Query**); reduced rework by standardizing loading/error/empty states and form validation behaviors across the product.
- Integrated frontend and backend through clean API contracts stable DTO payloads, consistent validation, and normalized error formats so the UI remained stable even while backend workflows evolved; improved developer efficiency by documenting endpoints and sharing Postman collections for QA and regression checks.
- Delivered backend services with **ASP.NET Core Web APIs** for core HR workflows (balances, eligibility calculations, approvals) while using **Node.js/Express** for supporting services and integrations (async/event-driven patterns), keeping responsibilities clear and minimizing cross-service coupling.
- Implemented secure workflows using **JWT/OAuth concepts** and RBAC patterns; designed role-aware UI behavior that matched backend authorization, reducing permission-related defects and preventing users from attempting restricted actions.
- Improved reliability and delivery speed by tightening CI/CD using **GitHub Actions** (lint/test/build checks), Dockerized dev environments, and consistent environment configuration; reduced “works in dev but fails in test” issues by aligning local/test/prod behavior.
- Built and integrated a **GenAI chatbot experience** for employee/self-service support using RAG patterns document ingestion, embeddings indexing, vector retrieval, and grounded response generation (LangChain/LlamaIndex concepts);

collaborated with stakeholders to define safe-answer boundaries and shipped monitoring/evaluation hooks to track retrieval quality.
- Mentored teammates through PR reviews and pairing sessions, focusing on practical React and API integration patterns that improved code readability and reduced repeated bugs across feature teams.

ICICI Home Finance Company (Contract) - Software Developer
Noida, India | Feb 2022 – Jun 2023
Domain: Financial Services / Reporting & Workflow Systems | **Stack:** React, TypeScript, C#, ASP.NET Core Web API, SQL Server, REST, Azure, Docker, Git-based CI/CD

- Built internal finance tools where data correctness and traceability mattered daily configuration screens, reporting dashboards, workflow experiences partnering with stakeholders to clarify business rules early and reduce sprint churn caused by late requirement changes.
- Developed React + TypeScript dashboards with complex filtering, drilldowns, and search; improved usability for analysts by keeping UI state predictable and consistent across routes and refreshes, reducing confusion during heavy reporting workflows.
- Implemented and maintained **ASP.NET Core APIs** with clean separation of concerns (controller/service/data layers) and defensive validation; enabled faster feature additions by keeping business rules centralized and minimizing duplicated logic across endpoints.
- Tuned SQL Server performance by analyzing execution plans, rewriting heavy joins/aggregations, and adding indexes aligned to real filter patterns; reduced slow report loads and timeouts during peak usage windows.
- Worked in an Azure-first environment where release readiness mattered; supported deployment verification, environment configuration checks, and production troubleshooting using logs/telemetry while balancing delivery speed with system stability.
- Improved engineering discipline through Git-based workflows, PR reviews, and CI checks; collaborated closely with QA to build strong test scenarios, reducing integration defects and improving release confidence.

Multiplier Solutions — Software Developer / Full Stack Developer
Hyderabad, India | Jan 2019 – Jan 2022

Domain: Healthcare Portals & Case Management | **Stack:** C#, ASP.NET MVC/Web API, React (select modules), JavaScript, SQL Server, Docker, AWS

- Delivered healthcare applications used by clinical and support staff to manage cases, tasks, and follow-ups; built support-first workflows where clarity and error resistance directly impacted day-to-day operations and patient-facing turnaround time.
- Built backend features in **ASP.NET MVC/Web API** using service-layer patterns to keep business rules consistent across screens and endpoints; improved maintainability by separating validation, business logic, and data access so changes were safer as requirements evolved.
- Modernized key UI modules with React to improve form usability, client-side validation, and error messaging; reduced data-entry mistakes by guiding users through clearer workflows and preventing invalid submissions earlier in the process.
- Designed and maintained **SQL Server schemas**, stored procedures, and constraints with a focus on data integrity and traceability; implemented audit-friendly history tracking patterns to support compliance reviews and internal investigations.
- Standardized environments using Docker and supported AWS delivery patterns; partnered with QA and domain SMEs during walkthroughs/UAT, and handled production issues via logs and DB traces, delivering low-risk fixes without destabilizing adjacent modules.
- Practiced early production discipline: handled tickets/incidents, reproduced edge cases, and documented fixes, which improved team confidence and reduced repeated defects in high-traffic modules.

PROJECTS
Leave Management & Research Support Platform — Multi-Tier .NET Application (Enterprise + SaaS)
Tech Stack: React, Next.js, TypeScript, C#, ASP.NET Core Web API, SQL Server/PostgreSQL, Entity Framework Core, REST/GraphQL, Redis, Docker, GitHub Actions, AWS
I built a workflow-driven platform that helps employees and HR teams manage leave requests, approvals, balances, eligibility rules, and policy enforcement. On the frontend, I implemented React + TypeScript screens for employee/manager/admin workflows using reusable components, consistent UI states (loading/error/empty), and performance-first patterns like lazy loading and code splitting so data-heavy pages stayed responsive. On the backend, I designed ASP.NET Core Web APIs using layered architecture (controllers → services → EF Core/data access), keeping rules centralized and easier to test and extend as policies changed. I modeled accrual/balance rules in the database and reduced export/report timeouts through indexing and execution-plan-driven query tuning. For delivery, I used Docker for environment consistency and GitHub Actions CI/CD to automate build/test steps,

making releases repeatable and safer.

GenAI HR Policy Assistant — RAG Chatbot + Observability (LFG/Perky)
Tech Stack: React/Next.js, TypeScript, Node.js, Python tooling (for evaluation/scripts), LangChain/LlamaIndex, vector DB concepts (embeddings indexing + retrieval), AWS (S3-style storage concepts), Docker, CI/CD
I built an HR self-service assistant that answers leave/policy questions using a retrieval-first approach instead of guessing. I created a document ingestion flow (policies/FAQs), chunked and embedded content, indexed it into a vector store, and implemented retrieval so the chatbot could ground responses in the most relevant policy sections before generating an answer. On the product side, I integrated the assistant into a React/Next.js chat UI with follow-ups and safe fallback behavior when confidence was low (route to HR/support instead of hallucinating). To make it production ready, I added logging for retrieval traces, response outputs, and failure cases, and I supported lightweight evaluation workflows to improve prompts and retrieval quality over time. Dockerized deployments and CI/CD kept the feature stable and repeatable across environments.

Financial Analytics Admin & Reporting Console
Tech Stack: React, TypeScript, C#, ASP.NET Core Web API, SQL Server, REST, Azure, Docker, Git-based CI/CD
This internal console helped finance teams configure thresholds and reporting rules, then explore analytics results with filters, drilldowns, and search. I built the React UI with a component-first approach so new dashboards could be added quickly without breaking existing screens. On the backend, I delivered .NET APIs with stable response contracts and defensive validation so UI and QA could depend on predictable behavior. I spent significant time tuning SQL Server queries for large datasets—rewriting joins/aggregations, adding indexes aligned with real filter usage, and validating improvements using execution plans—so analysts could load reports reliably during peak usage. I collaborated with stakeholders to refine reporting definitions and with QA to ensure deployments remained stable in Azure environments.

Healthcare Case Management Web App
Tech Stack: ASP.NET MVC/Web API, React (select modules), JavaScript, SQL Server, Docker, AWS
I worked on a case management system used by clinicians and support teams to track cases, tasks, and communications in one place. On the backend, I implemented workflows that enforced consistent case lifecycle rules and ensured critical actions were traceable for review. On the UI side, I modernized select modules with React and improved validation and error messaging so staff could complete tasks faster with fewer mistakes. I maintained stored procedures and schema constraints to protect integrity and added audit-friendly history tracking for compliance. I worked directly with clinical SMEs and QA through walkthroughs/UAT to match real-world workflows and supported production issues through logs/DB traces and low-risk fixes.

EDUCATION

Master of Science – Computer Science | Avila University, Kansas, USA | Dec 2024 | GPA: 3.63 / 4.00
Bachelor of Engineering – Computer Engineering | Kakinada Institute of Engineering and Technology, India | May 2020 | GPA: 3.58 / 4.00

CERTIFICATIONS & RECOGNITION

ASP.NET Professional Certificate – Udemy
Frontend Developer – Udemy (JavaScript / React)
Prize Winner – NVIDIA Developer Contest (2024) – Generative AI application